MIL-N-63400 (AR)
AMENDMENT 2
05 AUGUST 1988
SUPERSEDING
AMENDMENT 1
19 December 1983

MILITARY SPECIFICATION

NETWORKS, CAPACITOR - RESISTOR

This amendment forms a part of Military Specification MIL-N-63400 (AR), dated 19 October 1981, and is approved for use within the U.S. Army Armament, Munitions and Chemical command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Preamble, lines 1 and 2: Delete “U.S. Army ... command,” and substitute "U.S. Army Armament Research, Development and Engineering Center (ARDEC),”.

PAGE 2

DRAWINGS:

U.S. . . . COMMAND: Delete “COMMAND” and substitute “CENTER”.

Line 4: Add “9287175 - Network, Capacitor - Resistor”.

PAGE 4

Thermal shock, Column 4: Delete "± 0.5" and Substitute "+ 0.7”.

Power and Voltage Conditioning, Column 4: Delete "± 0.5" and substitute "+ 0.7”.

Resistor - Temperature Characteristic, Column 4: Delete "+ 250* and substitute "+ 300”.

Power Rating: Add “, Voltage Rating”:

Column 2, herein: Add “3.4.3, 3.4.2”;

Column 5, Capacitor Elements: Add "+ 10".

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Line 25, Columns 1, 3, 4, 5, 6: Add to the Columns “High Temperature Exposure, 3.25, ± 1.0, -, percent Change”.

NOTES: Add "2/ For inaccessible resistor or capacitor elements: see 4.5.2 and 4.5.3 herein.”

PAGE 5

3.4.1: Add the following sentence: “The changes in resistance and capacitance for the combined Thermal Shock and Power and Voltage Conditioning Tests, between the initial and final measurements shall not exceed ± (0.7 percent + 0.01 ohm) for resistor elements and ± 10 percent for capacitor elements.”

3.4.2 Delete "Maximum voltage." and substitute "Voltage rating."

Add this sentence: “The rated voltage for the capacitor is 25 volts minimum.”

Add" “3.4.3 Power rating. The power rating at 70°C shall be 0.1 Watt for each resistor element and 1.4 Watts maximum for the total package. The power ratings at 25°c for 1000 hours of life shall be 0.125 Watt for each resistor element and 1.75 Watts maximum for the total package.”

PAGE 7

Table II:

Dielectric . . capacitor, Column 5: Following “4.6.14.2” add ", *4.5.3*.

Dissipation Factor, Column 5: Following "4.6.7” add “, "4.5.3".

Insulation resistance of the capacitor, Column 5: Following "4.6.15.2* add ", *4.5.3”.

PAGE 8

Group IV:

Dielectric withstanding voltage: Add “ (Network)”.

Insulation resistance: Add “(Network)”.

Group VII, Columns 1, 4, 5: Add to the columns "High temperature exposure, 3.4.1, 4.6.21".

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PAGE 9

4.4.2.1:

Group A inspection: Delete sentence and substitute "Group A inspection shall consist of the examinations and test specified in 4.4.2.4 and, unless noted otherwise, shall be made on the same set of sample units in the order shown."

a. Sampling plan. lines 1 and 2: Delete "subgroups 1 and 2" and substitute "subgroup 2".

PAGE 10

Group A: Delete "0.40% AQL" and substitute "1OO%" for each of the following: Thermal shock; Power and Voltage conditioning; DC resistance; Capacitance.

Delete in their entirety the following tests: Dielectric withstanding voltage of the capacitor; Dissipation Factor; Insulation resistance of the capacitor.

PAGE 11

4.4.2.5:

Dielectric withstanding voltage of the capacitor: Following "4.6.14.2" add ", *4.5.3".

Dissipator factor" Following "4.6.7" add ", *4.5.3".

Insulation resistance of the capacitor: Following "4.6.15.2" add ", *4.5.3".

NOTES: Add "*Method paragraph herein."

PAGE 12

Group C, Monthly: Delete "Short-time overload in its entirety".

Group C, subgroup 1, Quarterly: Delete "Dielectric withstanding voltage (network) and Insulation resistance (network) in their entirety".

Group C, subgroup 2, Semiannually: Delet "Solderability in its entirety".

Subgroup 3, Columns 1, 4, 5: Add to the columns the following "High temperature exposure, 3.4.1, 4.6.2.1".

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PAGE 13

Add the following paragraphs:

"4.5.2 Inaccessible resistor or capacitor elements. Where resistor or capacitor elements are not directly and independently available to terminal pins for testing, the test or measurement shall be made in the most direct manner between two (2) pins. The pin to pin measurement shall be used directly to determine the acceptability to section 3 requirement. Where voltage application is restricted because of parallel elements the voltage shall be limited to the rated voltage on any effected element".

"4.5.3 Tests for dissipation factor, dielectric withstanding voltage of capacitor. Dissipation factor, dielectric withstanding voltage and insulation resistance of the capacitors shall be measured at each point in First Article Inspection, Quality Conformance Inspection whenever a dielectric Withstanding voltage or insulation resistance is in the circuit in parallel to the capacitor the dissipation factor and insulation resistance tests of the capacitor are not required".

PAGE 14

6.3, line 4: Delete "Command and DRDAR-QAA-I" and substitute 'Engineering Center, ATTN: AMSMC-QAT-I (D)".

6.4, line 4: Delete "Command and DRDAR-QAA-M" and substitute "Engineering Center, ATTN: AMSMC-QAT-M (D)".

6.5, line 3: Delete "Command and (ARRADCOM)" and substitute "Engineering Center (ARDEC)".

line 4: Delete "as," and substitute 'as U.S. Army Armament Research and Development Command (ARRADCOM) ,";

line 7: Delete "ARRADCOM" and substitute "ARDEC".

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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